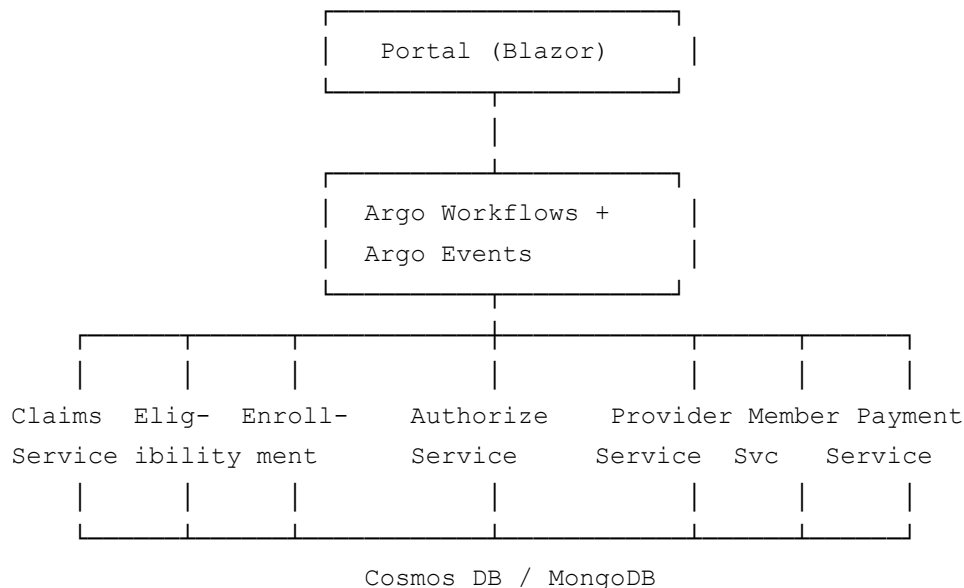


Cloud Health Office

Open-Source Core Administration Processing System (CAPS) for Health Plans

A cloud-native, multi-tenant payer platform with FHIR R4 APIs, X12 EDI processing, claims scrubbing, and CMS-0057-F compliance built on .NET 8, Kubernetes, and Argo Workflows.

Architecture



16 microservices covering claims, eligibility, enrollment, authorization, attachments, appeals, member, provider, payment, benefit plans, coverage, reference data, sponsors, tenants, and trading partners. See [Architecture](#).

Key Capabilities

CMS-0057-F FHIR R4 APIs — Patient Access, Provider Access, Payer-to-Payer, Prior Auth, CDS Hooks. [OpenAPI specs](#) and [compliance quickstart](#).

X12 EDI Processing — 270/271 eligibility, 275 attachments, 276/277 claim status, 278 prior auth, 834 enrollment, 835 remittance, 837P/I/D claims. Bidirectional X12-to-FHIR

mapping.

Claims Scrubbing — 30+ pre-adjudication validation rules with provider pre-edit suggestions. Providers see payer edits before submitting. [Claims scrubbing quickstart](#).

Multi-Tenant SaaS — Tenant isolation, self-service signup, Stripe billing, per-tenant rule configuration.

Argo Workflows — Kubernetes-native orchestration for X12 file processing, claims pipelines, and tenant onboarding.

Quick Start

```
git clone https://github.com/aureliaware/cloudhealthoffice.git
cd cloudhealthoffice
docker-compose up -d
```

Portal: <http://localhost:5000> **FHIR Metadata:**

<http://localhost:3000/fhir/r4/metadata> **Claims Scrubbing:**

<http://localhost:3100/api/v1/claims/validate>

See [Quickstart Guide](#) for detailed setup.

Project Structure

cloudhealthoffice/	
├─ api/	OpenAPI specs, quickstarts, Postman collection
├─ config/	Configuration schemas and examples
├─ containers/	Container images (x12-parser, sftp-fetcher, etc.)
├─ core/	Shared type definitions and validation
├─ docs/	All documentation (architecture, deployment, security
├─ infrastructure/	Argo, K8s, Helm, Azure Bicep, Kafka, monitoring
├─ schemas/	X12 XSD schemas
├─ scripts/	Setup, deploy, testing, migration scripts
├─ services/	.NET microservices (16 bounded contexts)
├─ site/	Marketing website
├─ src/	FHIR APIs, Portal, AI engine, Security, Functions
├─ tests/	Unit, integration, E2E tests + fixtures
└─ tools/	Migration wizard

Technology Stack

- **.NET 8 / TypeScript** — Microservices and FHIR APIs
 - **Blazor Server** — Portal UI
 - **Cosmos DB / MongoDB** — Multi-tenant data
 - **Kubernetes + Argo Workflows** — Orchestration
 - **Kafka** — Event streaming
 - **Azure** — Cloud deployment (Bicep IaC)
 - **SMART on FHIR / OAuth 2.0** — Authentication
 - **Stripe** — SaaS billing
-

Documentation

Topic	Link
Architecture	docs/architecture/ARCHITECTURE.md
CMS-0057-F Compliance	api/quickstarts/cms-0057f-compliance-quickstart.md
Claims Scrubbing	api/quickstarts/claims-scrubbing-quickstart.md
Quickstart	docs/onboarding/QUICKSTART.md
Deployment	docs/deployment/DEPLOYMENT.md
Security	SECURITY.md
API Reference	api/openapi/
Roadmap	docs/roadmap/ROADMAP.md

Related Projects

- [Cloud Dental Office](#) — Provider-side dental practice management (Blazor, .NET 8, microservices)

- Together, CHO + CDO provide complete provider-to-payer interoperability
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